WIRELESS CONTROLLED PICK AND PLACE ROBOTIC ARM

EC5078 – ROBOTICS

A PROJECT REPORT

Submitted by

PREM ARUN P	**2022504012**
DHIVAKAR PCK	**2022504034**
VARUN G	**2022504041**
SIRANJEEVI M	**2022504045**
ABISHEKRAGUL S	**2022504051**

in partial fulfillment for the award of the degree

of

BACHELOR OF ENGINEERING

IN

ELECTRONICS AND COMMUNICATION ENGINEERING

MADRAS INSTITUTE OF TECHNOLOGY

ANNA UNIVERSITY CHENNAI 600 044

DECEMBER - 2024

ANNA UNIVERSITY CHENNAI 600 044

BONAFIDE CERTIFICATE

Certified that this project report **"WIRELESS CONTROLLED PICK AND PLACE ROBOTIC ARM"** is the Bonafide work of **"PREM ARUN P (2022504012), DHIVAKAR PCK (2022504034), VARUN G (20225040401), SIRANJEEVI M (2022504045), ABISHEKRAGUL S (2022504051)"** who carried out the project work under my supervision.

SIGNATURE

Dr. L RAJESH

SUPERVISOR

Teaching Fellow ,

Department of Electronics Engineering,

Madras Institute of Technology,

Anna University,

Chennai – 600044.

ACKNOWLEDGEMENT

We consider it as our privilege and our primary duty to express our gratitude and respect to all those who guided and inspired us in the successful completion of the project.

With profound gratitude, we sincerely acknowledge with thanks the opportunity provided to us by our respectful **Dr.K.RAVICHANDRAN**, Dean, Madras Institute of technology, Anna University for providing good environment and facilities.

We thank our respectful Head of the Department **Dr.D.MEGANATHAN,** Professor and Head of Electronics Engineering, Madras Institute of technology, Anna University for his encouragement during the Project work.

We sincerely express our gratitude to our guide **Dr.L.RAJESH,** Teaching Fellow , Department of Electronics Engineering for his valuable guidance throughout the course of this project.

Place Chennai

Date 24-12-2024

PREM ARUN P (2022504012)

DHIVAKAR PCK (2022504034)

VARUN G (2022504041)

SIRANJEEVI M (2022504045)

ABISHEKRAGUL (2022504051)

ABSTRACT

The “Pick and Place Robotic Arm Manipulator” is a compact and efficient robotic system designed for handling small, lightweight objects. This project focuses on developing a 3 Degree of Freedom (DOF) robotic arm manipulator controlled via a Bluetooth module (HC-05). The system leverages wireless communication to enable seamless control, offering flexibility and ease of operation.

The robotic arm is equipped with four independent joints, providing the capability to perform basic movements such as gripping, lifting, rotating, and positioning small objects with precision. The Bluetooth module allows real- time control of the manipulator through a mobile application or a dedicated controller, eliminating the need for physical connections and enhancing user convenience.

Due to its lightweight design, this robotic arm is ideal for applications in educational demonstrations, research, and small-scale automation tasks. It showcases the integration of mechanical design, embedded systems, and wireless communication to create a cost-effective and reliable solution for scenarios requiring precise handling of lightweight objects.

CONTENTS

S.NO	TITLE	PG.NO
	ABSTRACT	**IV**
1	INTRODUCTION	***1***
2	COMPONENTS USED	
	2.1 ARDUINO UNO	***2***
	2.2 SG90 SERVO MOTORS	***4***
	2.3 HC-05 BLUETOOTH MODULE	***5***
	2.4 MEDIUM DENSITY FIBRE BOARD	***7***
3	METHODOLOGY	
	3.1 COMPONENT SELECTION	***8***
	3.2 SYSTEM DESIGN AND ASSEMBLY DISPLAY AND USER INTERACTION	***8***
	3.3 SOFTWARE DEVELOPMENT AND TESTING	***9***
	3.4 IMPLEMENTATION AND OPTIMIZATION	***9***
4	RESULTS	***9***
5	CONCLUSION	***10***
6	REFERENCES	***10***

CHAPTER 1

INTRODUCTION

The “Pick and Place Robotic Arm Manipulator” is an innovative and versatile system designed to perform tasks involving the precise handling of small, lightweight objects. Robotic arms have come an integral part of automation in various fields, including manufacturing, education, and research, due to their ability to perform repetitive tasks efficiently and with high precision.

This project focuses on developing a robotic arm with 3 Degrees of Freedom (DOF) that can be controlled wirelessly using a Bluetooth module (HC- 05). The system utilizes servo motors for joint articulation, base rotation, and gripper operation, enabling smooth and accurate movements. The wireless control is achieved through a mobile application or Bluetooth-enabled device, providing an intuitive and user-friendly interface for real-time operation.

The project highlights the integration of mechanical design, embedded systems, and wireless communication to create a compact and cost-effective solution for small-scale automation tasks. Its applications range from educational demonstrations to prototype testing, showcasing the potential of robotics in simplifying everyday tasks and enhancing productivity.

By leveraging advanced technologies and a systematic design approach, this project aims to provide a hands-on understanding of robotic systems while contributing to the growing field of automation and robotics.

CHAPTER 2

COMPONENTS USED

2.1 ARDUINO UNO

Fig (i) – Arduino UNO

The Arduino Uno is a versatile microcontroller board based on the ATmega328P, widely used in embedded systems and robotics. Known for its simplicity, open-source nature, and ease of use, it is ideal for both beginners and professionals to prototype electronic projects.

Technical Specifications

Microcontroller

- **Model:** ATmega328P
- **Architecture:** 8-bit AVR RISC
- **Clock Speed:** 16 MHz

Power Supply

- **Operating Voltage:** 5V

- **Input Voltage:** (Recommended) 7-12V
- **Input Voltage:** (Limit) 6-20V
- **DC Current per I/O Pin:** 20 mA (max)
- **DC Current for 3.3V Pin:** 50 mA

Memory

- **Flash Memory:** 32 KB (0.5 KB used by bootloader)
- **SRAM:** 2 KB
- **EEPROM:** 1 KB

Input/Output Pins

- **Digital I/O Pins:** 14 (6 with PWM output)
- **Analog Input Pins:** 6 (10-bit resolution)
- **PWM Channels:** 6

Connectivity

- **USB:** Type-C for programming and communication
- **Serial Communication:** UART, SPI, I2C/TWI
- **LED Indicators:** Power, TX, RX, Built-in LED on pin 13

Other Features

- Reset Button for manual reset
- Voltage Regulator for stable 5V and 3.3V outputs
- **Dimensions:** 68.6 mm × 53.4 mm
- **Weight:** 25 g

Applications in this Project

1. **Central Controller:** Manages all components and executes control algorithms.
2. **Servo Motor Control:** Generates PWM signals for precise servo movements.
3. **Bluetooth Interface:** Connects to the HC-05 module for wireless command reception.
4. **Signal Processing:** Translates Bluetooth signals into movement instructions.
5. **Real-Time Control:** Ensures synchronized servo operation for accurate tasks.
6. **Power Distribution:** Efficiently powers servos and the Bluetooth module.
7. **Ease of Programming:** Facilitates development and debugging via Arduino IDE and libraries.

2.2 SG90 Servo Motors

Fig (ii)-SG90 Servo Motors

The SG90 is a compact and lightweight servo motor widely used in robotics and automation projects.

Electrical Specifications

- **Operating Voltage:** 4.8V to 6.0V
- **Operating Current:** 100 mA to 250 mA (depending on load)
- **Control Signal:** PWM (Pulse Width Modulation)
- **Pulse Width Range:** 500 µs to 2400 µs
- **PWM Frequency:** 50 Hz

Mechanical Specifications

- **Torque:** 1.8 kg·cm (at 4.8V)
- **Speed:** 0.12 sec/60° (at 4.8V)
- **Angle Range:** 0° to 180°
- **Gear Type:** Plastic
- **Weight:** 9 g

Pin Description

1. **Brown Wire (Ground)**
    - Connects to the ground (GND) of the power supply or microcontroller, providing the reference voltage for the motor.
    - Connection GND of Arduino, Raspberry Pi, or external power supply.
2. **Red Wire (Power/Vcc)**
    - Connects to the power supply, providing the operating voltage for the motor.
    - Voltage Range 4.8V to 6.0V.
    - Connection 5V pin of Arduino, Raspberry Pi, or external power source.
3. **Orange Wire (Signal)**
    - Receives the PWM signal from the microcontroller to control the motor's position. The PWM duty cycle determines the angular position of the shaft.
    - Connection PWM pin of Arduino, Raspberry Pi, or any microcontroller.

Applications in this Project

- Enable smooth joint movements, base rotation, vertical lifting, and gripper operation for handling small, lightweight objects.
- Provide precise control for the arm's 3 Degrees of Freedom (DOF), ensuring accurate and real-time execution of pick-and-place tasks essential for automation and precision.

2.3 HC-05 BLUETOOTH MODULE

Fig (iii)-HC-05 Bluetooth Module

The HC-05 is a Bluetooth module designed for short-distance wireless communication, widely used in IoT and embedded systems to enable data exchange without physical connections.

Key Features

- **Bluetooth Version:** 2.0+EDR (Enhanced Data Rate)
- **Frequency:** Operates at 2.4 GHz ISM band
- **Range:** Up to 10 meters (33 feet)
- **Modes:** Master or slave mode
- **Power Supply:** 3.3V to 6V (optimal at 5V)
- **Communication Protocol:** UART (Universal Asynchronous Receiver-Transmitter)
- **Default Baud Rate:** 9600 bps (adjustable via AT commands)
- **Status Indicators:** LED shows connection status (blinking for pairing mode, steady for successful connection)

Pin Description

- **VCC:** Power supply (3.3V–5V)
- **GND:** Ground connection
- **TXD:** Transmits data (connects to RXD of microcontroller)
- **RXD:** Receives data (connects to TXD of microcontroller)
- **EN:** Enables the module (used for entering AT command mode)

Applications in this Project

1. **Wireless Control:** Enables communication between the robotic arm and a mobile device without physical connections.
2. **Real-Time Command Transmission:** Allows precise and responsive control of the robotic arm.
3. **Low-Cost and Reliable Communication:** Provides an affordable and dependable solution for Bluetooth connectivity in embedded systems.

2.4 MEDIUM DENSITY FIBRE BOARD (MDF)

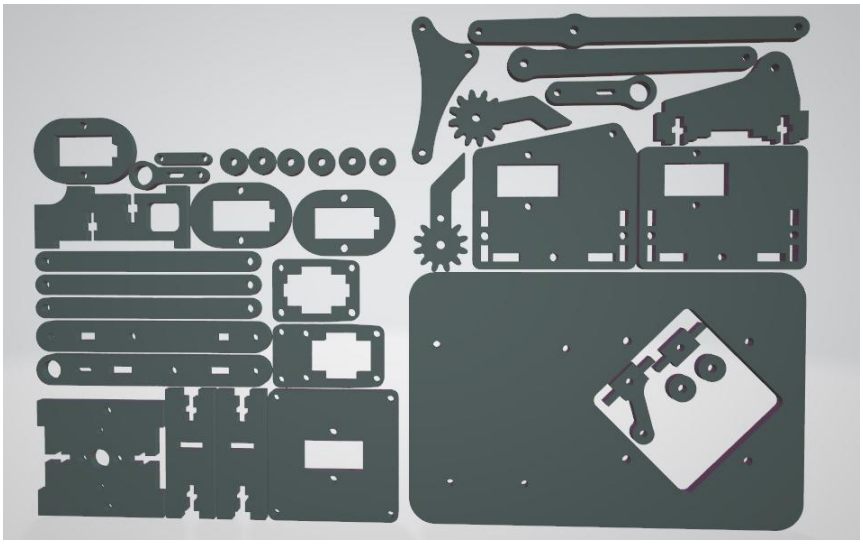


Fig (iv) – MDF board

Medium Density Fiber board is a versatile engineered wood product made from wood fibres, wax, and resin. It offers a smooth, uniform surface, making it ideal for precise cutting and shaping in various applications, including the construction of robotic arm frames.

Key Features

1. **Smooth Surface:** Provides a flat, even surface, ideal for precise cutting and finishing.
2. **Durability:** Strong and resistant to wear, making it suitable for structural applications.
3. **Workability:** Easy to machine, drill, and shape, ideal for custom designs.
4. **Lightweight:** Lighter than solid wood, making it easier to handle and assemble.
5. **Cost-Effective:** More affordable than other hardwood materials while offering similar strength.

Application in this Project

- **Frame Construction:** Used for building the structural components of the robotic arm.
- **Precision Cutting:** Smooth surface enabled accurate and clean cuts for parts.
- **Lightweight:** Reduced the overall weight of the arm while maintaining strength.
- **Durability:** Ensured stability and long-lasting performance of the arm.
- **Easy Machining:** Facilitated custom design adjustments for joints and gripper.

CHAPTER 3

METHODOLOGY

3.1 Component Selection

Hardware Components

- Arduino Uno Central controller for servo motors and Bluetooth communication.
- Servo Motors (SG90) Provide movement for joints and gripper.
- HC-05 Bluetooth Module Enables wireless control from a mobile device.
- Power Source 9V DC battery to power the Arduino and servos.
- Mechanical Frame and Gripper Constructed using MDF (Medium Density Fiber) board , designed with AutoCAD for precision and assembly.

Software Tools

- Arduino IDE For programming motor control and Bluetooth communication.
- Bluetooth Controller App To send commands from a mobile device.

3.2 System Design and Assembly

Mechanical Design

- Robotic arm with 3 Degrees of Freedom (DOF), powered by servo motors.
- Servo-driven gripper for object manipulation.
- MDF Board used for the arm's frame, with parts designed in AutoCAD for precise dimensions.

Circuit Design

- Servo Motors Connected to PWM-enabled pins (D9, D10, D11) on Arduino.
- HC-05 Bluetooth Module TX to RX and RX to TX of Arduino for communication.
- Power Supply Arduino powered via USB or 9V adapter.

Assembly

- Mount servo motors and attach the gripper to the end effector.
- Connect components as per the circuit design.

3.3 Software Development and Testing

Arduino Programming

- Program to receive Bluetooth commands and control servo angles.

Bluetooth Controller App

- Pair HC-05 with a mobile device to send control commands.

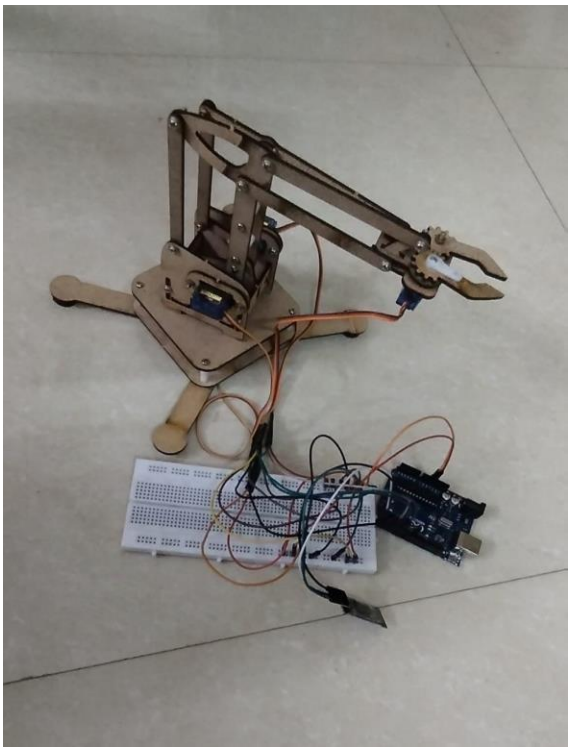
Testing and Calibration

- Test each servo motor's response and adjust angles for smooth movement.
- Calibrate the gripper for secure object handling.

3.4 Implementation and Optimization

- Perform pick and place tasks by rotating the base, adjusting the middle joint, and activating the gripper.
- Troubleshoot issues like servo jitter by debugging code and recalibrating angles.
- Optimize for smoother and faster performance.

RESULT



The pick-and-place robotic arm was assembled using a lightweight frame, with servo motors mounted at each joint to provide the necessary degrees of freedom. The gripper was attached to the end effector and calibrated for handling lightweight objects.

The Arduino Uno controlled the system, receiving commands from the HC-05 Bluetooth module and generating PWM signals for the servos. The Bluetooth module was paired with a mobile device for wireless control via an app. The system was powered by an external 9V adapter.

The arm successfully performed pick-and-place tasks, with smooth joint movements and precise gripper operation. Minor adjustments were made during testing to optimize performance.

CONCLUSION

The pick-and-place robotic arm successfully handles small objects with precision using servo motors for joint movement, a gripper for object handling, and the HC-05 Bluetooth module for wireless control. The Arduino Uno ensures smooth operation of the system.

This project demonstrates the feasibility of using affordable components for functional automation and provides a foundation for future improvements, such as adding more degrees of freedom, increasing payload capacity, or integrating sensors for advanced features. It serves as a stepping stone toward broader automation applications.

REFERENCES

1. **"Arduino Robotics"** by John-David Warren, Josh Adams, and Harald Molle. This book provides guidance on building robotic systems using Arduino, including robotic arm designs and control systems.

2. **"The MeArm: A Pocket-Sized Robotic Arm"** on Autodesk Instructables Website. This tutorial provides a step-by-step guide to building a small, hackable robotic arm, similar to your project.